

Geo-spatial modelling of vehicle crime in Northern Ireland using computer vision to identify environmental factors.

Actuarial Research Conference 2025

James Hannon^{1,2}, Dr. Adrian O'Hagan^{2,3}, Prof. Declan French⁴, and Prof. Barry Quinn⁵.

¹Centre for Research Training in Foundations of Data Science, University College Dublin,

²School of Mathematics and Statistics, University College Dublin,

³Insight Centre for Data Analytics, University College Dublin,

⁴Queen's Business School, Queen's University Belfast,

⁵Ulster University Business School, Belfast

Email: james.hannon1@ucdconnect.ie

Table of contents

1. Motivation
2. Research Aim
3. Data
4. Methodology
5. Key Results
6. Conclusion

Motivation

Explaining Spatial Risk in Vehicle Crime Using the Built Environment

Spatial risk is modelled via **structured + unstructured random effects**

But: random effects don't explain **why** risk varies

Crime Prevention Through Environmental Design (CPTED) suggests physical features influence crime

1. Access control
2. Natural surveillance
3. Territorial reinforcement

Research Aim

Detect **environmental features** (e.g. CCTV, street lights) in **Google Street View (GSV)** images

Objective: Build an interpretable model of geographic crime risk

Insurance relevance: Demonstrate potential for enhancing models of location-based risk

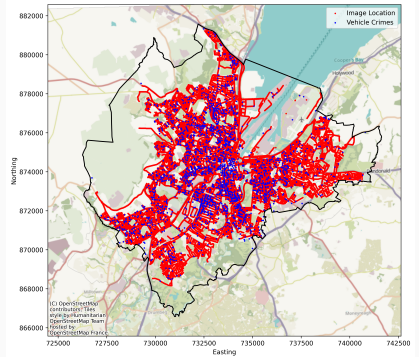
Data

- **Coverage:** 55,418 locations across Belfast
- **Image Set:** 6 images per location (360° views) → 332,508 total
- **Timeframe:** 96% captured between 2019–2023
- **Seasonality:** Mostly late spring to early autumn
- **Limitations:** Some areas excluded (e.g. airports, private roads); occasional blurring



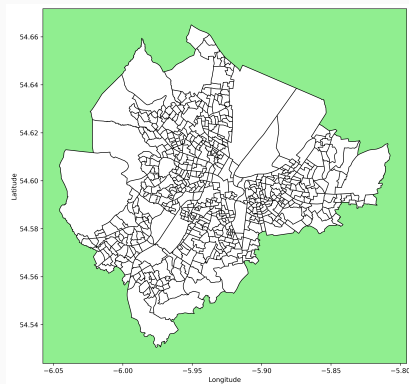
Vehicle Crime

- **Source:** data.police.uk
- **Location:** Belfast
- **Timeframe:** January 2019 – July 2023
- **Incidents:** 4,316 recorded vehicle crimes
- **Crime Types:**
 - Theft from vehicle
 - Theft of vehicle
 - Vehicle interference



Census

- Based on the 2021 Northern Ireland Census
- Data published at the Data Zone level: 3,780 zones across Northern Ireland
- Zones designed to be:
 - Population-balanced
 - Socially homogeneous
 - Geographically compact
- **Belfast** includes:
 - 638 Data Zones (DZ)



Methodology

Object Detection

- **GroundingDINO** [2] is an open-set object detector that combines DINO [3], a transformer-based detector, with grounded pre-training.
- **Input:** Image and a prompt
- **Output:** Bounding boxes, object labels, Confidence scores
- **Performance:** 52.5 AP on COCO dataset

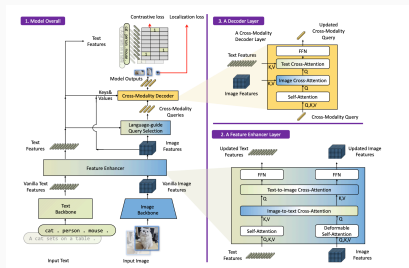


Table 1: Object categories and corresponding search terms used in Grounding DINO prompts.

Category	Prompt
Vehicle	Car, van, truck, bus, motorcycle
Graffiti	Graffiti
Litter	Litter, rubbish
Light	Light, Streetlight, lamp
CCTV	CCTV, camera
Boundary	Fence, wall, gate, rail
Window	Window
Building	Building, house, shop
Door	Door
Vegetation	Tree, hedge, grass, field

For each data zone we have information relating to

- Size
- Population
- Age demographics
- Distance travelled to work
- Health status
- Highest Qualification
- Employment
- Number of vehicles

Bayesian Model

We model the number of vehicle crimes y_i using a Poisson GLM with an offset, fixed effects, and spatial random effects:

$$y_i \sim \text{Poisson}(\lambda_i) \quad (1)$$

$$\log(\lambda_i) = \log(V_i) + \mathbf{x}_i^\top \beta + \sigma \left(\sqrt{\frac{\rho}{S}} \phi_i + \sqrt{1 - \rho} \theta_i \right) \quad (2)$$

Model components:

Offset: $\log(V_i)$

Fixed effects: $\mathbf{x}_i^\top \beta$

Spatial effects: $\sigma \left(\sqrt{\frac{\rho}{S}} \phi_i + \sqrt{1 - \rho} \theta_i \right)$

S is derived from the spatial graph $\mathcal{G}$ to ensure the structured spatial effect ϕ has approximately unit variance.

V_i is the number of vehicles in each data zone.

We use the regularised horseshoe prior for the fixed effects:

$$\beta_j \sim \mathcal{N}(0, \tilde{\lambda}_j^2 \tau^2) \quad (3)$$

$$\tilde{\lambda}_j^2 = \frac{c^2 \lambda_j^2}{c^2 + \tau^2 \lambda_j^2} \quad (4)$$

$$\lambda_j \sim C^+(0, 1) \quad (\text{local shrinkage}) \quad (5)$$

$$\tau \sim C^+(0, a^2) \quad (\text{global shrinkage}) \quad (6)$$

$$c^2 \sim \text{Inv-Gamma}(\alpha, \beta) \quad (\text{regularisation term}) \quad (7)$$

Hyperparameter specification:

- $a = \frac{p_0}{D - p_0} \cdot \frac{\sigma}{\sqrt{n}}$, where:
 - p_0 : prior guess for number of relevant covariates
 - D : total number of covariates
 - n : number of observations
 - We use $\sigma = 2$ (recommended by Piironen & Vehtari)
- We set $\alpha = 2, \beta = 0.1$

Besag-York-Mollié 2 (BYM2) Model

We use Penalised Complexity (PC) priors for the spatial component:

$$\sigma \left(\sqrt{\frac{\rho}{s}} \phi_i + \sqrt{1 - \rho} \theta_i \right)$$

$$\frac{1}{\sqrt{s}} \phi_i \mid \phi_{-i} \sim \mathcal{N} \left(\sum_{j=1}^{n_i} \phi_j, 1 \right) \quad (\text{structured, ICAR}) \quad (8)$$

$$\theta_i \sim \mathcal{N}(0, 1) \quad (\text{unstructured}) \quad (9)$$

$$\sigma \sim \text{Exp}(b) \quad (\text{marginal SD}) \quad (10)$$

$$\rho \sim \text{PC prior} \quad (\text{mixing parameter}) \quad (11)$$

We choose:

- $b = -\frac{\log(\alpha)}{U}$ to satisfy $\mathbb{P}(\sigma > U) = \alpha$
- $U = \frac{0.5}{0.31}$ and $\alpha = 0.01 \rightarrow$ gives β marginal SD of 0.5

For ρ , the PC prior is derived from the spatial graph $\mathcal{G}$:

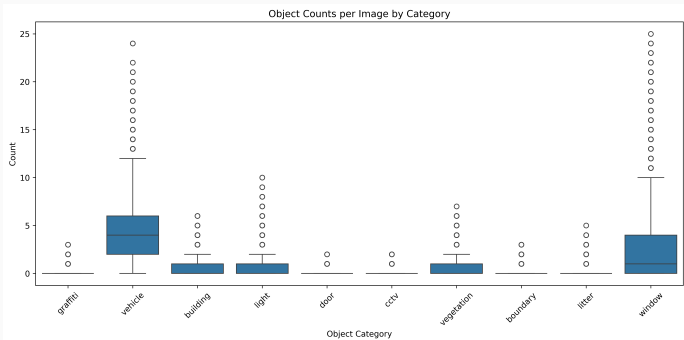
- Set so that $\mathbb{P}(\rho < U) = \alpha$
- We use $U = 0.5$ and $\alpha = \frac{2}{3}$

Model Estimation and Evaluation

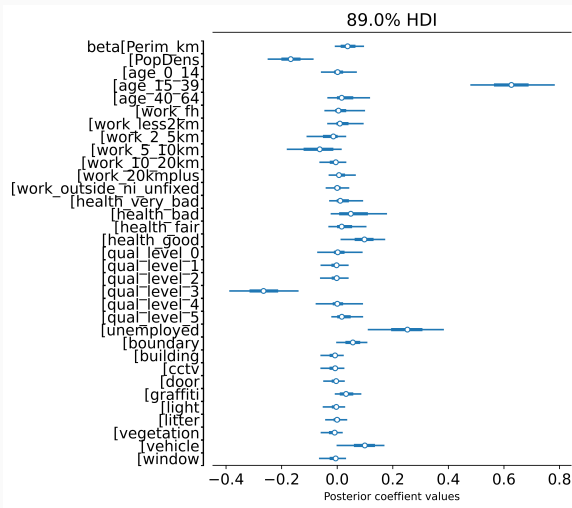
- Model fitted in **PyMC** [1] using NUTS (No-U-Turn Sampler; adaptive HMC)
- 4 chains with 5,000 draws each (1,000 burn-in)
- Convergence diagnostics: $\hat{R}$, effective sample size (ESS), Monte Carlo standard error (MCSE), and trace plots
- Inference based on 89% highest density intervals (HDI)
- Model fit assessed using posterior predictive checks and spatial residual maps
- Model comparison via expected log pointwise predictive density (ELPD), estimated using Pareto-smoothed importance sampling LOO-CV:
 - Fixed effects only
 - Spatial effects only
 - Both fixed and spatial effects

Key Results

GroundingDINO output



Influential Fixed Effects

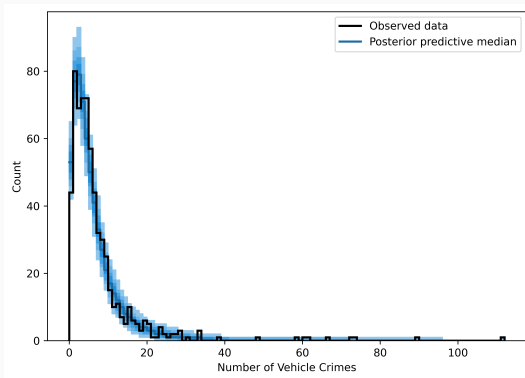


$$\sigma((\sqrt{\rho/s})\phi_i + (\sqrt{1-\rho})\theta_i)$$

	mean	sd	89% HDI	MCSE mean	MCSE sd	ESS bulk	ESS tail	$\hat{R}$
σ	0.75	0.04	[0.69, 0.82]	0.001	0.000	2355	5710	1
ρ	0.73	0.09	[0.59, 0.89]	0.003	0.001	1124	2358	1

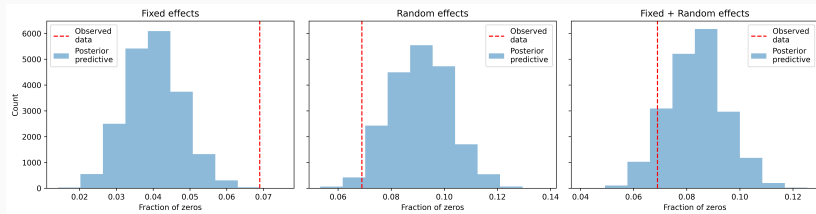
- The **structured spatial effect** (ϕ) accounts for **25%** of the variation
- The **unstructured spatial effect** (θ) accounts for **45%**
- The **fixed effects** explain **16%**
- The remaining **14%** is due to **covariation between components**

Posterior Predictions



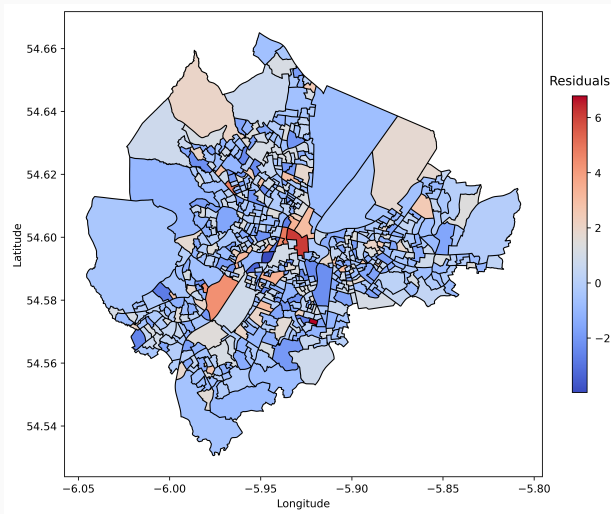
Rank	Model	ELPD LOO	p LOO	Δ ELPD	w	se	dse
1	Fixed and Spatial Effects	-1612.7	350.9	0.0	0.85	22.2	0.0
2	Spatial Effects only	-1638.4	385.4	25.7	0.15	21.2	9.0
3	Fixed Effects only	-2489.3	249.9	876.6	0.0	179	168.8

Zero-Inflation and dispersion



	Fixed Effects only	Spatial Effects only	Fixed and Spatial Effects
Dispersion Statistic	9.26	0.47	0.56

Residuals



Conclusion

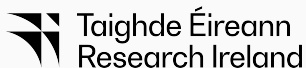
Summary

- **Built environment features**, such as graffiti and CCTV, can be automatically detected in GSV images and meaningfully linked to vehicle crime.
- **Bayesian spatial modelling** (BYM2 + fixed effects) captures spatial variation in crime while remaining interpretable.
- **Influential predictors** include:
 - Census variables: unemployment, proportion of young people
 - Detected features: number of vehicles, graffiti
- **Explained variation:** Fixed effects: 16% Spatial effects: 70% (structured + unstructured)
- **Best model performance** achieved with combined fixed + spatial effects:
 - Superior predictive accuracy (ELPD-LOO)
 - Fewer effective parameters than spatial-only model
 - Strong handling of zero-inflation and dispersion

Future improvements: account for covariate correlation using *Projection Predictive Variable Selection*

Acknowledgements

This publication has emanated from research jointly funded by Taighde Éireann – Research Ireland under Grant number 18/CRT/6049 and 12/RC/2289_P2, and the British Academy (BA)/Leverhulme Small Research Grants under Grant Number SRG2223/231522.



Questions?

References



O. Abril-Pla, V. Andreani, C. Carroll, L. Dong, C. J. Fonnesbeck, M. Kochurov, R. Kumar, J. Lao, C. C. Luhmann, O. A. Martin, et al.
Pymc: a modern, and comprehensive probabilistic programming framework in python.

PeerJ Computer Science, 9:e1516, 2023.



S. Liu et al.

Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection.

arXiv preprint arXiv:2303.05499, 2023.



H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum.

Dino: Detr with improved denoising anchor boxes for end-to-end object detection.

arXiv preprint arXiv:2203.03605, 2022.

Backup Slides

Label	Highest level of qualifications
qual_level_0	No qualifications
qual_level_1	1 to 4 GCSEs, O levels, CSEs (any grades); 1 AS Level; NVQ level 1; or equivalent
qual_level_2	5+ GCSEs (A*-C or 9-4), O levels (passes), CSEs (grade 1); 1 A level, 2-3 AS Levels; NVQ level 2, BTEC General, City and Guilds Craft; or equivalent
qual_level_3	2+ A Levels, 4+ AS Levels; NVQ Level 3, BTEC National, OND, ONC, City and Guilds Advanced Craft; or equivalent
qual_level_4	Degree (BA, BSc), foundation degree, NVQ Level 4+, HND, HNC, professional qualifications (e.g., teaching, nursing); or equivalent
qual_level_5	Apprenticeships; other qualifications, equivalent unknown

Backup Slides

Calculating s : For a given adjacency structure (neighborhood matrix), the precision matrix $\mathbf{Q}$ is defined as:

$$\mathbf{Q} = \begin{cases} n_i & \text{if } i = j \\ -1 & \text{if } i \sim j \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

where n_i is the number of neighbours for area i . In practice, we use $\mathbf{Q}^-$, the generalised inverse of $\mathbf{Q}$. We let,

$$s^2 = \exp\left(\frac{1}{n} \sum_{i=1}^n \log(\mathbf{Q}_{ii}^-)\right) \quad (13)$$

where Σ is the covariance matrix.

Backup Slides

$\rho \sim PC$: Let $\mathbf{u}$ be a n -dimensional standardised Gaussian model with zero mean and precision matrix $\mathbf{R} > 0$, $\mathbf{v}$ be an independent zero mean random effects with unit variance $\mathcal{N}(\mathbf{0}, \mathbf{I})$, and where the mixing parameter ρ satisfies $0 \leq \rho \leq 1$. The more flexible model is $\sqrt{1-\rho}\mathbf{v} + \sqrt{\rho}\mathbf{u}$, and the base model is $\mathbf{v}$ (i.e., the model is flexible model when $\rho = 0$). Let $\Sigma_0 = \mathbf{I}$ and $\Sigma_1(\rho) = (1-\rho)\mathbf{I} + \rho\mathbf{R}^{-1}$, then

$$2\text{KLD}(\rho) = \text{tr}(\Sigma_1(\rho) - n - \log(|\Sigma_1(\rho)|)) \quad (14)$$

$$= n\rho\left(\frac{1}{n}\text{tr}(\mathbf{R}^{-1}) - 1\right) - \log(|(1-\rho)\mathbf{I} + \rho\mathbf{R}^{-1}|) \quad (15)$$

and $d(\rho) = \sqrt{2\text{KLD}(\rho)}$

$$\text{ELPD} = \sum_{i=1}^n \mathbb{E}_{p(\theta|X)}[\log(p(y_i|\theta))] \quad (16)$$